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a) $K_2^3 K_3 / K_1$ b) $K_1 K_3^3 / K_2$

c) $K_2 K_3^3 / K_1$ d) $K_2 K_3 / K_1$

25. A 20 litre container at 400 K contains CO_2 (g) at pressure 0.4 atm and an excess of SrO (neglect the volume of solid SrO). The volume of the container is now decreased by moving the movable piston fitted in the container. The maximum volume of the container, when pressure of CO_2 attains its maximum value will be :

Given that : $SrCO_3$ (S) ⇌ SrO (S) + CO_2(g)

K_P = 1.6 atm (NEET 2017)

a) 2 litre b) 5 litre

c) 10 litre d) 4 litre

26. If there is no change in concentration, why is the equilibrium state considered dynamic ?

27. For a given reaction at a particular temperature, the equilibrium constant has constant value. Is the value of Q also constant ? Explain.

28. What the relation between K_P and K_C. Give one example for which K_P is equal to K_C.

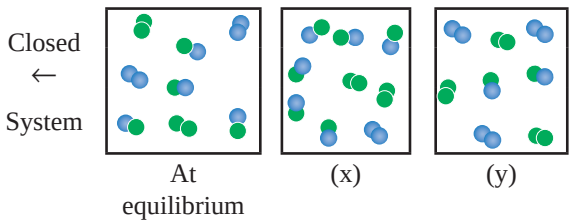
29. For a gaseous homogeneous reaction at equilibrium, number of moles of products are greater than the number of moles of reactants. Is K_C is larger or smaller than K_P.

30. When the numerical value of the reaction quotient (Q) is greater than the equilibrium constant (K), in which direction does the reaction proceed to reach equilibrium ?

31. For the reaction,

$A_2(g) + B_2(g) \rightleftharpoons 2AB(g)$; ΔH is –ve.

the following molecular scenes represent different reaction mixture (A – green, B – blue)



i) Calculate the equilibrium constant K_P and (K_C).

ii) For the reaction mixture represented by scene (x), (y) the reaction proceed in which directions ?

iii) What is the effect of increase in pressure for the mixture at equilibrium.

32. State Le-Chatelier principle.

33. Consider the following reactions,

a) $H_2(g) + I_2(g) \rightleftharpoons 2\,HI$

b) $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

c) $S(s) + 3F_2(g) \rightleftharpoons SF_6(g)$

In each of the above reaction find out whether you have to increase (or) decrease the volume to increase the yield of the product.

34. State law of mass action.

35. Explain how will you predict the direction of a equilibrium reaction.

36. Derive a general expression for the equilibrium constant K_P and K_C for the reaction